

CSC 317/2504: Computer Graphics

Course web site (includes all course information):

<https://github.com/panuelosj/computer-graphics-csc317-F26>

	LEC 0101/2001	LEC 0201
Instructor	Prof. Karan Singh	Jonathan Panuelos
Contact	karan@dgp.toronto.edu	jonathan.panuelos@mail.utoronto.ca
Lecture	Tue 13:00–15:00 MS2170	Tue 15:00–17:00 MS2172
Tutorial	Thu 13:00–14:00 UC140	Thu 15:00–16:00 BR200
Office Hours	TBD	TBD

TAs

Vishnu Nittoor, Karran Pandey, Yunze Wei, Sophia Yang, Yue Chang, Yi Xin Lok. csc317tas@cs.toronto.edu

Questions:

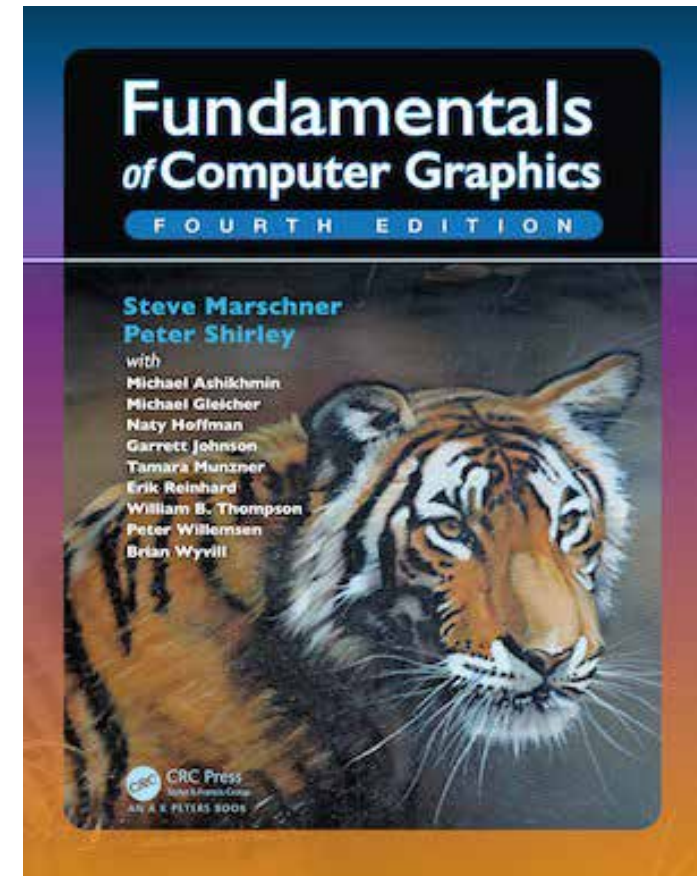
on Assignments: Github issues pages

on Administrative stuff: Quercus <https://q.utoronto.ca/courses/441943>

Textbooks: Fundamentals of Computer Graphics

You will need:

- A computer to install and program the assignments.
- An internet connection for online material and to join any zoom meetings (unlikely).
- Availability during class and tutorial times.
- The textbook for required readings



Schedule (from the webpage)

Week	Sunday	Mon (Quiz Release)	Tue (Lectures)	Thu (Tutorial/Test)	Fri (Due + Release)
1	Sep 6	Labour Day	Introduction + Rasterization	Tut 1	a1 + a2 released
2	Sep 13	q1 released	Ray Casting + Tracing	Tut 2	q1 due
3	Sep 20	q2 released	Ray Casting + Tracing	Tut 3	a1 + q2 due · a3 released
4	Sep 27		Ray Casting + Tracing	Test 1 (Rasterization–Ray Tracing)	
5	Oct 4		BVH	Tut 4	a2 due · a4 released
6	Oct 11	Thanksgiving · q3 released	Meshes + Shaders + Animation	Tut 5	q3 due
7	Oct 18	q4 released	Meshes + Shaders + Animation	Tut 6	a3 + q4 due · a5 released
8	Oct 25	Reading Week	Reading Week	Reading Week	Reading Week
9	Nov 1		Meshes + Shaders + Animation	Test 2 (BVH–Shaders)	
10	Nov 8		Guest Lecture / New Work	Tut 7	a4 due · a6 released
11	Nov 15	q5 released	Kinematics (Drop Day Tue Nov 17)	Tut 8	q5 due
12	Nov 22		Springs	Tut 9	a5 due
13	Nov 29		Current Research	Test 3 (Kinematics–Springs)	
14	Dec 6		<i>no class</i>	<i>no class</i>	a6 due

Marking Scheme (from the webpage)

%	Item
5	Assignment 1 — Raster Images
40	Assignments 2–6 (8% each)
10	5 Quizzes (Online, 2% each)
15	Test 1 (Rasterization–Ray Tracing)
15	Test 2 (BVH–Shaders)
15	Test 3 (Kinematics–Springs)

Academic Honesty Policy is on the webpage and is mandatory reading!

Tutorials sessions will be invaluable for the assignments.

Prerequisites (pre-test)

Background in math, CS and programming is required for this course.

A quick self-monitored pre-test (15-20 mins should be enough to do it)

1. Show you the mathematical background expected and your readiness for this course.
2. Show you what you need to brush up on. Questions about these math operations will not be answered by either Professors or TAs, we expect you to know this stuff.

Discrete Math

Q1: What is the intersection of the interval $\mathcal{A} = [-1, 3]$ and $\mathcal{B} = [0, 4]$? $\mathcal{A} \cap \mathcal{B} =$

Q2: In a balanced binary tree with n leaf nodes, how many internal nodes are there?

Q3: In a balanced binary tree with n leaf nodes, how deep is the tree?
(i.e., number of nodes on a shortest path from root to leaf)

Q4: What is $5!$ (five factorial)?

Q5: How many different ways are there to choose k items from a set of n items?

Linear Algebra

Q6: What size is the result of multiplying a 2×3 matrix and a 3×4 matrix?

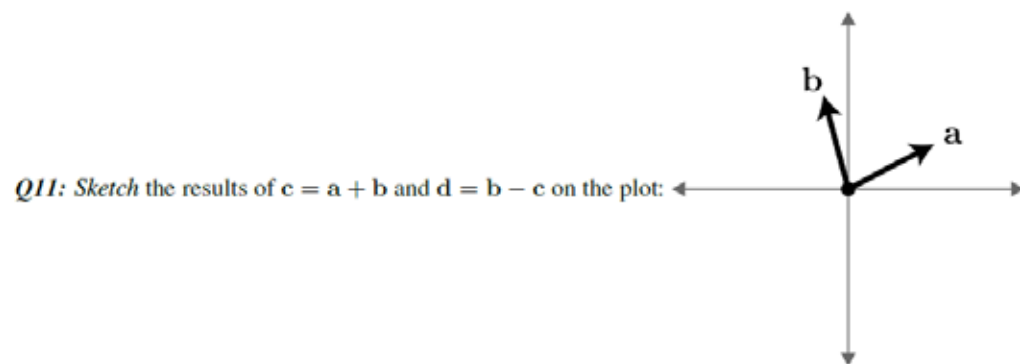
Q7: What is the dot product between the vector $\mathbf{a} = [1, 2, 3]$ and $\mathbf{b} = [4, 5, 6]$? $\mathbf{a} \cdot \mathbf{b} =$

Q8: What is the cross product between the vector $\mathbf{a} = [4, 0, 0]$ and $\mathbf{b} = [0, 0, -2]$? $\mathbf{a} \times \mathbf{b} =$

Q9: What is the matrix product of the matrix $\mathbf{A} = \begin{pmatrix} 1 & 0 & 4 \\ 0 & 2 & 1 \\ 0 & -3 & 0 \end{pmatrix}$ and the vector $\mathbf{b} = \begin{pmatrix} -4 \\ 5 \\ 6 \end{pmatrix}$?

$\mathbf{A} \mathbf{b} =$

Q10: Using matrices, express the solution to the following system of equations:
 $3x_1 + 2x_2 - 1x_3 = 10$, $2x_1 - 10x_3 = 0$, $-9x_1 + 9x_2 = -1$. (Form the system, but do not solve).



Mappings

Q12: Suppose we have a $f : \mathcal{A} \rightarrow \mathcal{B}$, what needs to be true to call f a *well-defined function*?

Q13: Is $f(x) = x^2$ invertible? Why (not)?

Q14: If x and y are real numbers, what sets are the domain and range of $f(x, y) = x^2 + y^2$? (use $\mathbb{R}$)

Logarithms

Q15: Express $\log(a^b)$ in terms of $\log a$ and b . $\log(a^b) =$

Q16: Express $\log(ab)$ in terms of $\log a$ and $\log b$. $\log(ab) =$

Trigonometry

Q17: Express $\cos(a + b)$ in terms of $\cos a$ and $\cos b$. $\cos(a + b) =$

Q18: Given a triangle with side lengths a, b, c and opposite angles A, B, C , express $\sin A$ in terms of a, b , and $\sin B$.
 $\sin A =$

Q19: Given a right triangle with acute angles A and B , express the following in terms of opposite side length a and b :
 $\sin A =$ $\cos A =$ $\tan A =$

Quadratic equation

Q20: How many real-value solutions for x are possible to an equation of the form $ax^2 + bx + c = 0$?
A) zero; B) one; C) two; D) three; E) zero, one or two; or F) zero, one, two or three.

Q21: Identify the solution(s), if any, to $x^2 - 4x - 5 = 0$.

Linear interpolation

Q22: What is *the* linear function $f : [0, 1] \rightarrow \mathbb{R}$, so that $f(0) = a$ and $f(1) = b$?

Q23: What is *the* linear *vector-valued* function $f : [0, 1] \rightarrow \mathbb{R}^3$, so that $f(0) = \mathbf{a}$ and $f(1) = \mathbf{b}$?

Derivatives

Q24: What is the derivative of the function $f(t) = 3t^2$? $\frac{df}{dt} =$

Q25: What is the second derivative of the function $f(t) = 3t^2$? $\frac{d^2f}{dt^2} =$

Q26: What is the partial derivative of the function $f(x, y) = 3x^2 - 3y^2$ with respect to y ? $\frac{df}{dy} =$

Q27: Given a function $f : \mathbb{R}^3 \rightarrow \mathbb{R}$, how many dimensions does its gradient have?

Q28: Using partial derivative and vector notations express the gradient of a function $f(x, y, z) : \mathbb{R}^3 \rightarrow \mathbb{R}$.
 $\nabla f =$

Geometry

Q29: What is the maximum number of intersections of a straight line and a pentagonal shape in 2D?

Q30: What is the volume of a sphere with radius r ?

Today

1. Introduction to Computer Graphics.
2. Preview of Assignments.
3. Raster Images.

Introduction to Computer Graphics

What is Computer Graphics?

Computers:

accept, process, transform and present information.

Computer Graphics:

accept, process, transform and present information
in a visual form.

Ok but... what is the course really about?

The science of turning the rules of geometry, motion and physics into (digital) pictures.

What its not about?

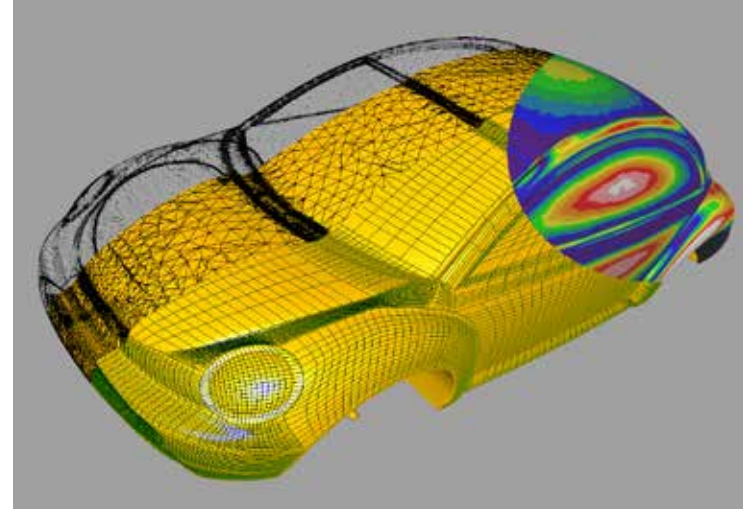
Photoshop, AutoCAD, Maya, Blender, Renderman, Graphics APIs.



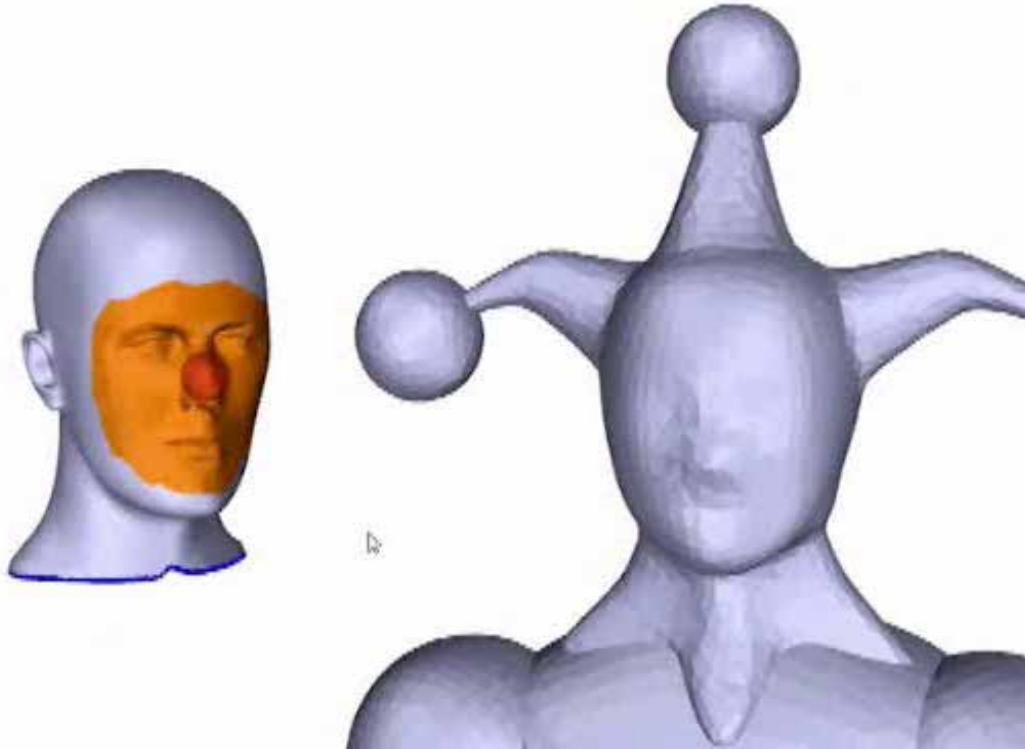
Evan Pan (CSC 317 alumnus) <https://evansamaa.github.io/>

“Core” Areas of Computer Graphics

- Form (modeling)
How do we represent (2D or 3D) objects & scenes?
How do we build these representations?
- Function, Behavior (animation)
How do we represent the way objects move?
How do we define & control their motion?
- Appearance (rendering)
How do we represent the appearance of objects?
How do we simulate the image-forming process?



Modeling



[MeshMixer. Schmidt, Singh, *SIGGRAPH 2010*]
www.meshmixer.com (acquired by Autodesk Inc.)

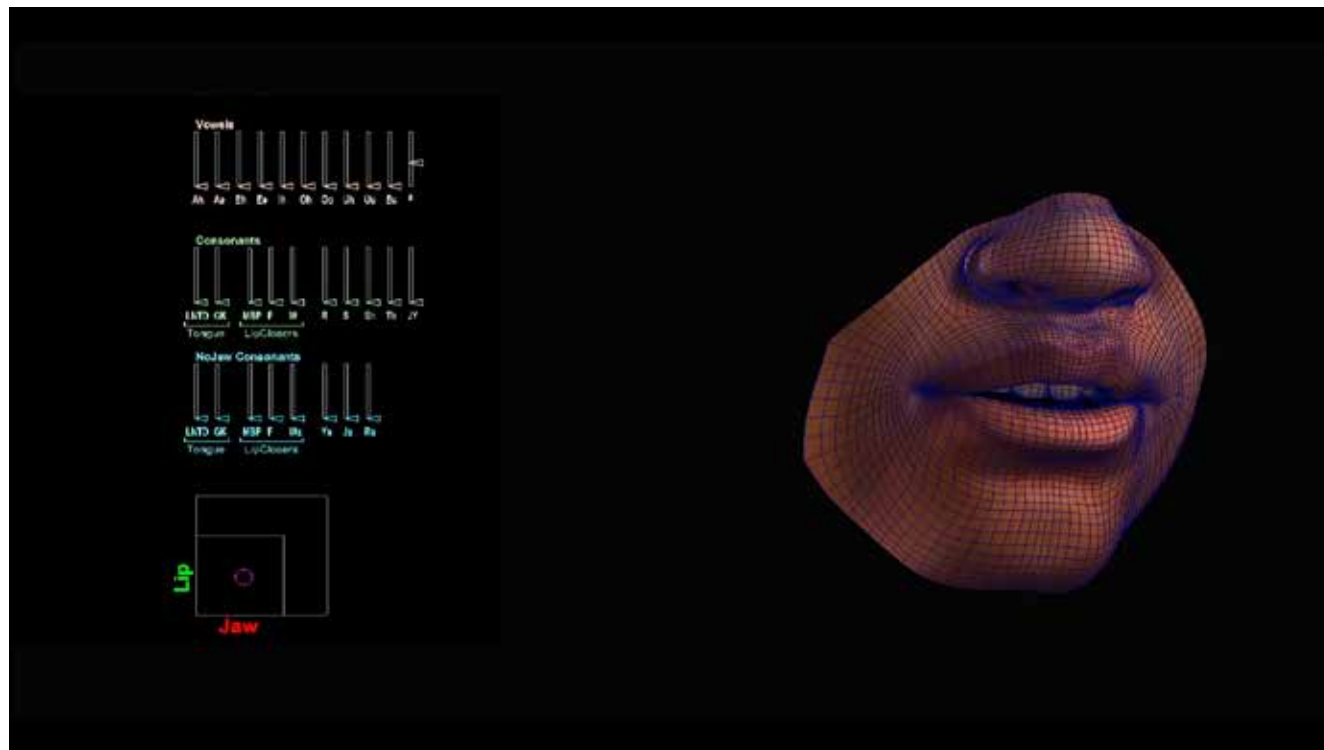


[Face-Extrusion Quadmeshes. Pandey, Baerentzen ,Singh, *SIGGRAPH 2022*]

Animation

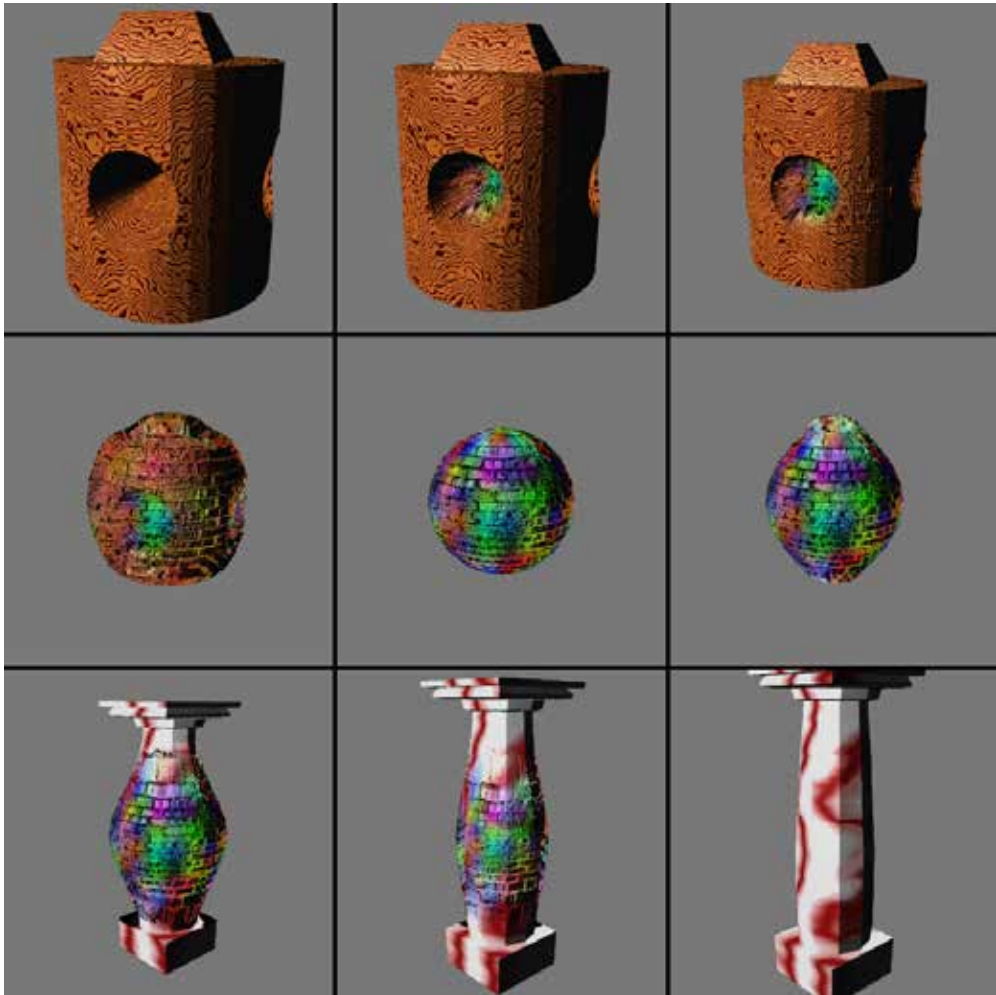


[Kinodynamic skinning using volume-preserving deformations, Angelidis, Singh, *ACM SCA 2007*]



[Visemenet: Zhou et al. *SIGGRAPH 2018*]

Rendering



[Singh, around 1992]



[Weta Digital 2022]

Other Areas of Computer Graphics

User Interaction

Virtual Reality

Visualization

Image Processing

3D Scanning and Fabrication

Computational Photography

Movies

Drive new directions in CG
Set quality standards for CG





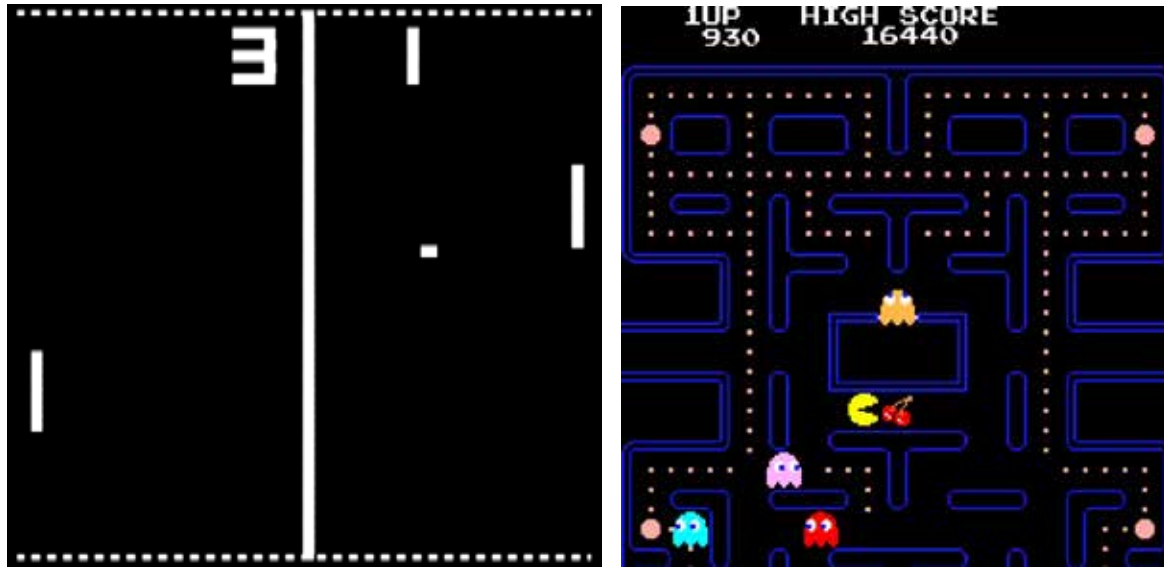
AVATAR

THE WAY OF WATER

Games

Drive interactivity and AI in CG

Push CG hardware to its limits





Stellar Blade
Shift Up

Jali

Assignment Previews

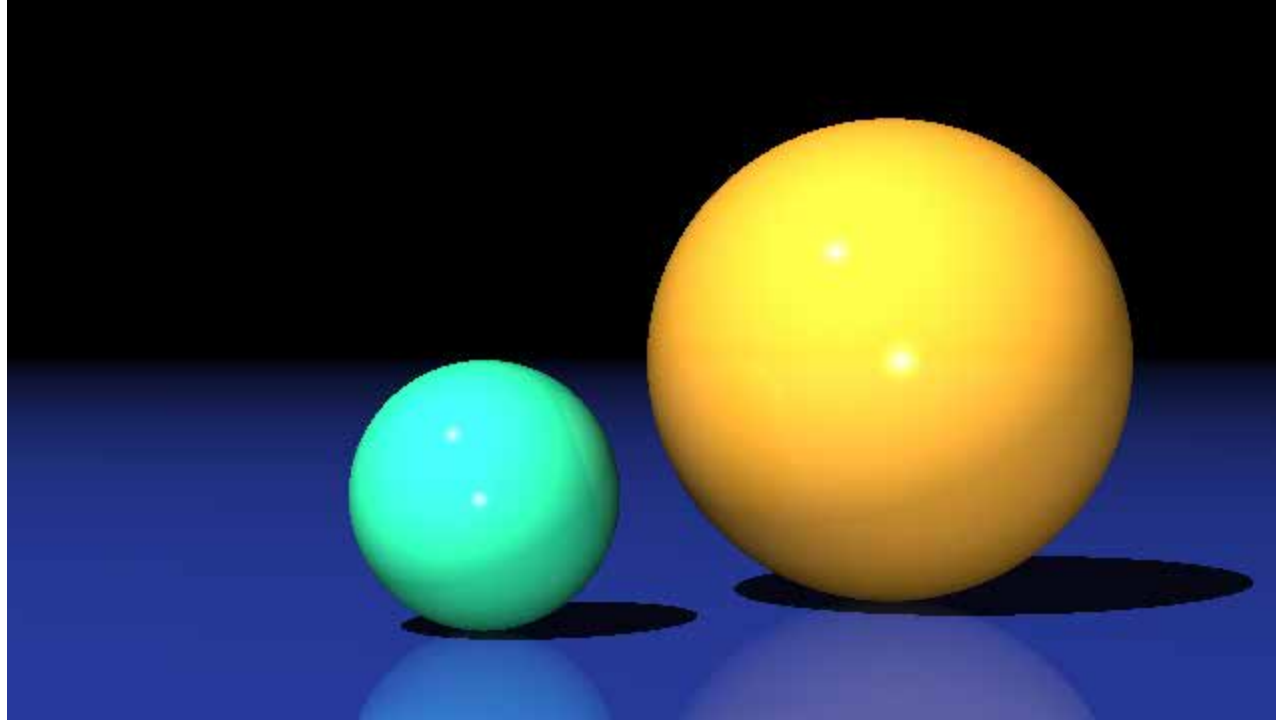
1. Raster Images
2. Ray Tracing
3. Boundary Volume Hierarchies
4. Meshes + Shaders + Animation
5. Kinematics
6. Mass-Springs

Be Creative!

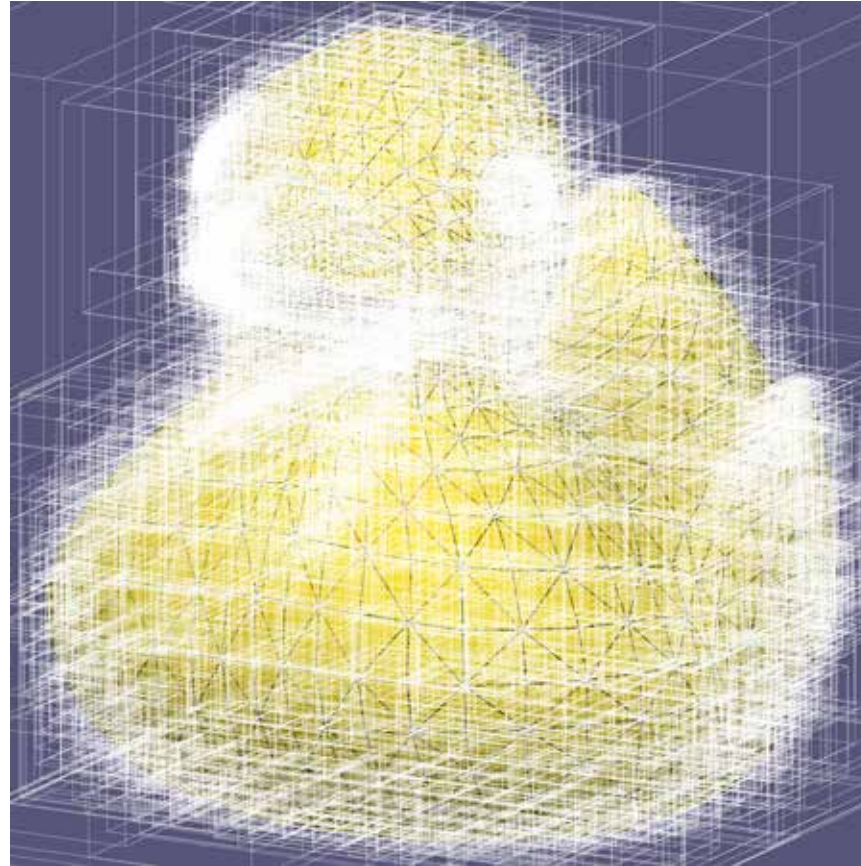
Raster Images



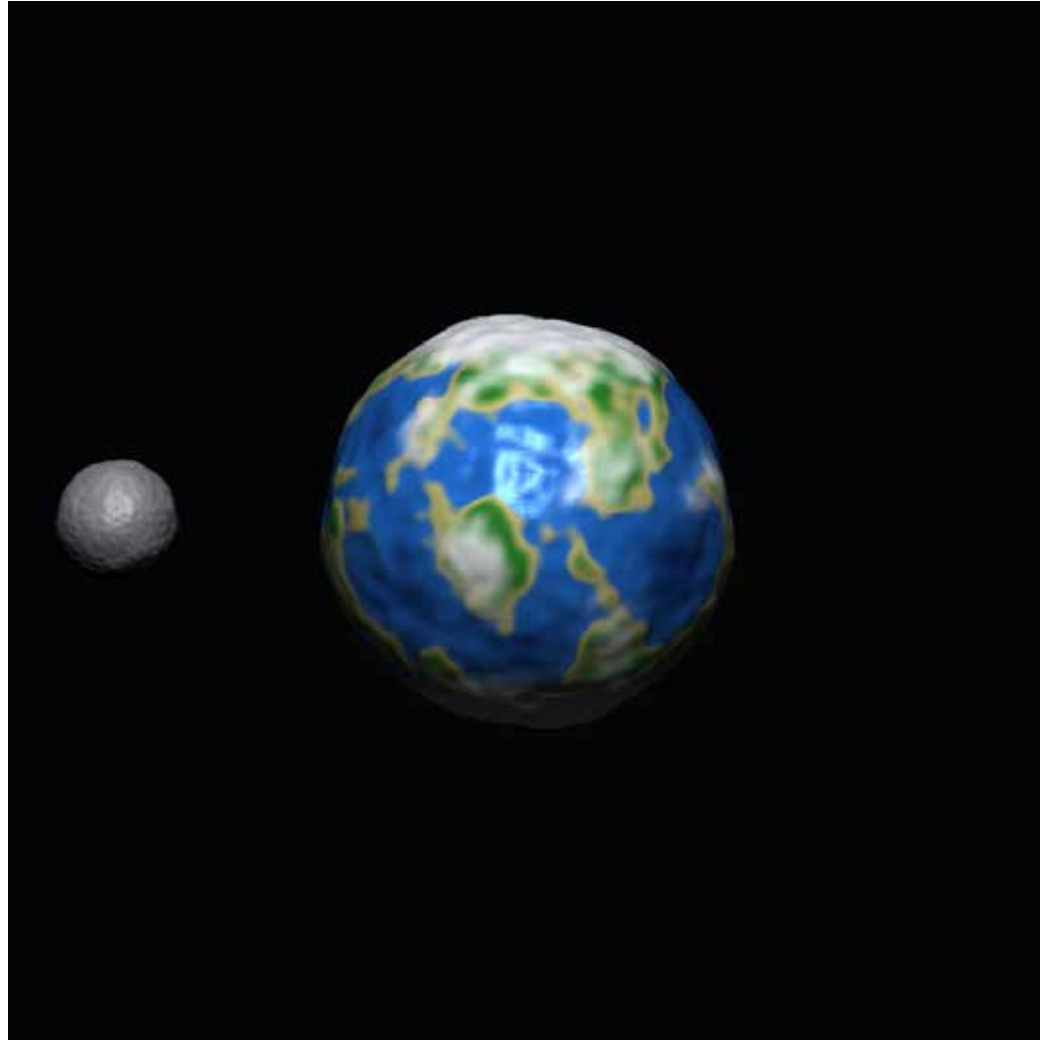
Ray Tracing



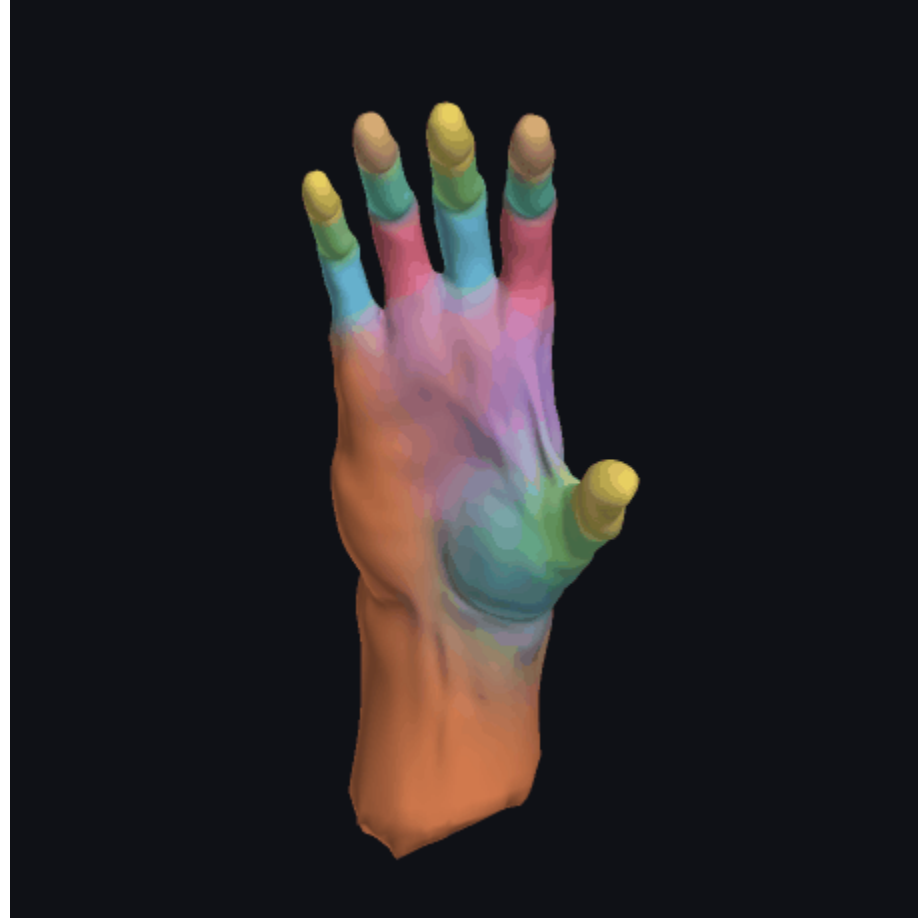
Boundary Volume Hierarchies



Meshes, Shaders and Animation



Kinematics



Mass-Springs



Next lecture >
Raster Images